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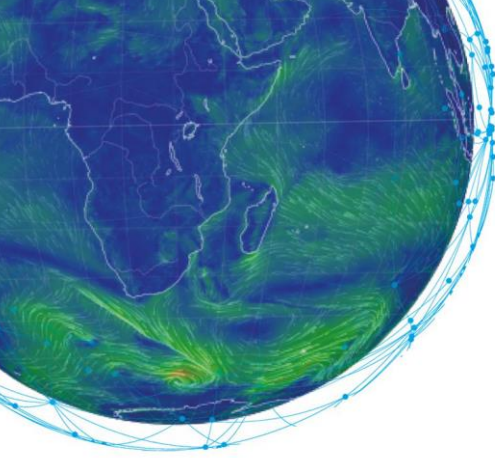
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Machine Learning in Weather & Climate

TRANSCRIPT

**The Digital Twins of Destination
Earth**

Expert: Peter Bauer



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The Digital Twins of Destination Earth

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Introduction

Hello and welcome to this podcast about The Digital Twins of Destination Earth. My name is Lisa Burke and today, I will interview Dr Peter Bauer, ECMWF director of Destination Earth (DestinE). He has been at ECMWF for more than 20 years, leading research on satellite data assimilation, model development and computing science. Dr. Bauer received his PhD in meteorology from the University of Hamburg, Germany. Before joining ECMWF, he occupied scientist positions at DLR in Germany, NOAA and NASA in the US, and IPSL in France.

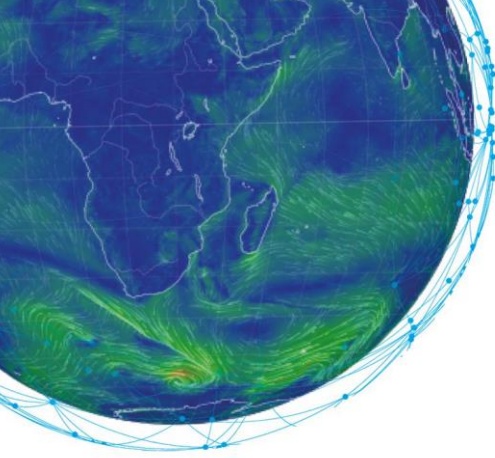
Podcast

Lisa: Hello Peter. So tell me, where does the Destination Earth initiative come from and where is it placed with respect to existing extremes and climate services?

Peter: The requirement for much advanced Earth-system simulation capabilities dates back to the Climate Modelling Summit that was organized at ECMWF in 2008 and where leading climate scientists called for a substantial upgrade in model spatial resolution as a necessary condition to overcome key errors in our climate predictions that are in the way of defining adaptation measures at regional and local level. Already then, the huge impact on the resulting computing and data handling footprint was recognized and even led to the proposition of a so-called CERN for climate.

The immediate predecessor of DestinE was a proposal for a 10-year, €1 billion flagship programme called ExtremeEarth that I coordinated in partnership with 18 of the leading institutes for Earth-system science, computational science and service providers in Europe. Already here, we created the link between much advanced simulation and observation capabilities enabled by supercomputing, and something really new that we called ExtremeEarth Science Cloud at the time. The latter was the predecessor of the digital twins in DestinE.

In the end, DestinE was successfully formulated and is now being funded from the new Digital Europe Programme of the European Commission that also funds the new generation of pre-exascale and exascale computers in Europe.



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Lisa: Can you tell us what digital twins are and how they add to what we can do with today's weather and climate prediction systems?

Peter: The term 'digital twin' stems from engineering where, for example, airplanes or complex factories were designed and operated with the help of computer simulations of these airplanes and factories. These are supported by real-time observations helping to monitor the operating environment and the status of important technical components. This framework allows to optimize design and operations in terms of efficiency, but also to predict potential component failures that could then be addressed before the incident occurs.

Translating this to the Earth system, we do some of this already in our present weather and climate prediction systems but there are critical elements that are missing.

First, an important step is to make models and the use of observations better so that the digital replica becomes more accurate. More realism will allow making better decisions.

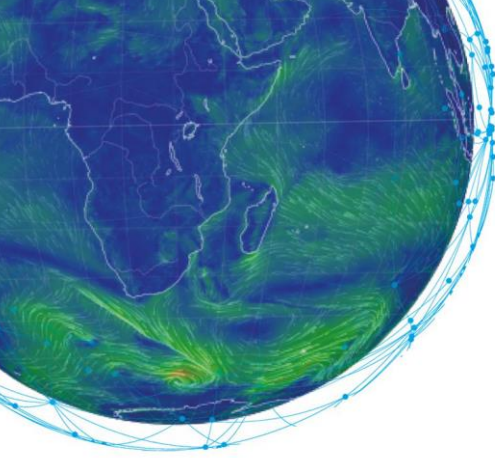
Second, while we need a better representation of the Earth-system physics in these systems, we need to make the step to application areas that matter for society: the management of water, energy and food resources, risks for health, lives and property. This translation requires the full integration of modelling and observation components for, for example, flooding, fires, pollution, irrigation and the connection to cost and risk modelling.

Third, we need an information system that allows users from all these sectors but also expert scientists to interact with models and data so that they can extract exactly the information they need for decision making. Digital twins require all three features coming together.

Lisa: Which technologies are the most important ones for making digital twins a reality?

Peter: Ideally, all digital technologies related to high-performance computing, extreme-scale (big) data handling, smart sensors and networks, the internet of things but also tools like machine and deep learning and mathematical methods. Cybersecurity will also play an important role as digital twins will access enormous data spaces and distributed workflows through the largest computing infrastructures. For all of those, security is becoming increasingly important.

One important aspect here is that none of these technologies has been designed for DestinE or society's adaptation to climate change. Our community therefore has to make major investments in creating the best-possible benefit from these general-purpose technologies for the specific purpose of climate change adaptation and for dealing with environmental extremes. This is by no means a plug-and-play exercise but needs substantial bespoke developments, for which we actually allocated a large part of our DestinE funding.



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Important are therefore also direct partnerships with companies developing these technologies. While most of the industrial decisions are made for maximizing profit, we may still be able to get their support for one of the most important challenge that our society is facing.

Lisa: Many digital twin publications and announcements highlight the importance of machine learning and artificial intelligence for digital twins. Is this also true for the Earth-system digital twins that you are developing in DestinE?

Peter: Indeed, machine and deep learning will support many aspects of DestinE, and we are only starting to tap into this resource now. While these methods are already decades old, the recent boost of both computing capabilities and the commercial push for exploitation created new training algorithms for vast amounts of data and a new universe of toolkits for a vast range of applications.

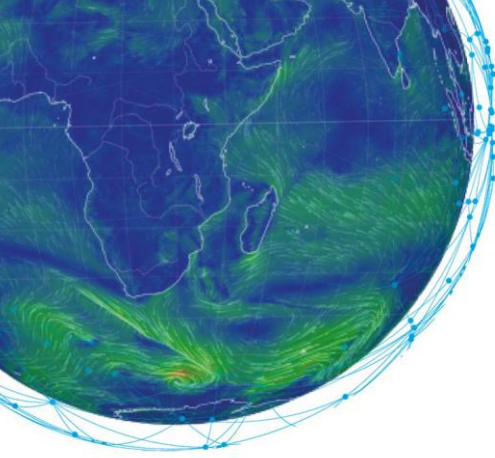
We are already developing machine-learning methods for maximizing the information content we draw from hundreds of millions of observations that we use every day, and which will drastically increase once commodity devices like mobile phones or the internet of things will be explored as well.

There is also a great potential for accelerating our complex model simulations on supercomputers through machine learning and to help extract user-specific information from the enormous amounts of data the digital twins will be generating. Particularly the above-mentioned translation of information from the physical to the human world will rely on machine learning as this does not necessarily follow mathematical equations and needs to be derived from data that is growing by the minute.

Lastly, the important topic of uncertainty quantification is where machine learning will be instrumental for the success of DestinE. Numerical models and satellite observations are not perfect, creating the link to the human world will not be perfect and letting a wide variety of users interact with such a complex system will create further uncertainties. Here, machine learning may be the only option for objectively quantifying and tracing the uncertainties along the production process and helping to make sensible decisions for our society.

Summary

Digital Twins of the Earth system offer new ways to combine the latest, highly accurate simulation models with observational data, link the physical system to sectors of societal relevance and create an interactive data space where users can fully immerse themselves in vast amounts of data and create their specific application domain. Creating these twins is a huge effort relying on the merging the latest science and digital technology developments. The European Destination Earth project will pave the way for this outstanding task, but needs global support to create such an information system for addressing one of the biggest challenges our society is facing.



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Thank you so much Peter for this great discussion. And for those who were listening to us and would like to discuss this subject in further detail, please use our Forum on Moodle. It will be a pleasure to hear from you.